From Atomic to Cosmic: A Summary of my work at the STFC

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MANTID CreateMonteCarloWorkspace

As Part of my placement in the Mantid team I was tasked to create an algorithm to test fitting functions written by scientists and developers. The algorithm is called *CreateMonteCarloWorkspace.* It generates a simulated workspace by sampling from the probability distribution of input data, in this case it uses a cumulative distribution function. Monte Carlo simulations work by sampling from a probability distribution. To do that, the algorithm:

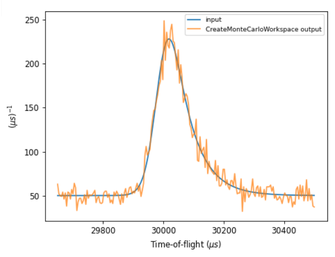
•Uses a **uniform random number** between 0 and 1.
•Maps that number to a **value (e.g., detector ID or bin)** using the **CDF**.

The input workspace's spectrum is interpreted as a **probability density function (PDF)**.
•The algorithm converts this to a **cumulative distribution**.
•It then randomly chooses events based on the weights given by this distribution.

Using a CDF guarantees:
•Correct proportions of events across spectra or bins.
•Reproducibility when given the same seed.
•Realistic modelling of experimental data.

```
/**
 *  Compute a normalized CDF [0..1] from the given histogram data.
 */
std::vector<double> CreateMonteCarloWorkspace::computeNormalizedCDF(const Mantid::HistogramData::HistogramY &yData) {
  std::vector<double> cdf(yData.size());
  std::partial_sum(yData.begin(), yData.end(), cdf.begin());
  double totalCounts = cdf.back();

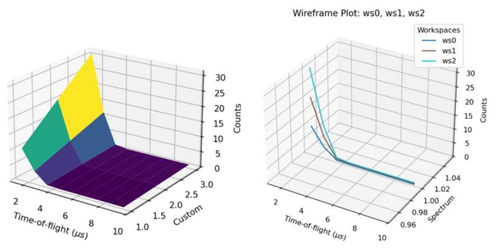
  if (totalCounts > 0.0) {
    // Normalize the CDF
    std::transform(cdf.begin(), cdf.end(), cdf.begin(), [totalCounts](double val) { return val / totalCounts; });
  } else {
    g_log.warning("Total counts are zero; normalization skipped.");
  }
  return cdf;
}
```



Noise generated from algorithm

Additional work on Mantid

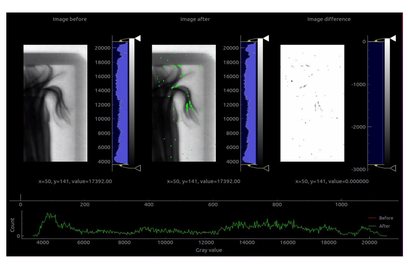
• Allow the Surface, Contour and Waterfall Plotting from Multiple Single-Spectrum Workspaces.
• Updated the entire code base from **Matplotlib** version 3.7 to version 3.9
• Harmonise workbench and mantid workbench entry points
• Resolved an issue where the graph window legend could be dragged outside the window and lost permanently
• Improved tabbing order across multiple windows for better navigation
• Fixed a malformed **XML** script that was causing the editor to crash



Multiple Single-Spectrum Plots

Mantid Imaging

As part of my time with Mantid Imaging, I introduced the ability for users to apply image processing operations to specific datasets rather than all loaded data. Added a feature to automatically reset brightness levels following flat fielding, ensuring consistent visual feedback and minimizing user intervention. Implemented a mechanism that allows users to cancel the loading of large or incorrect datasets mid-process, improving user control and responsiveness of the interface.

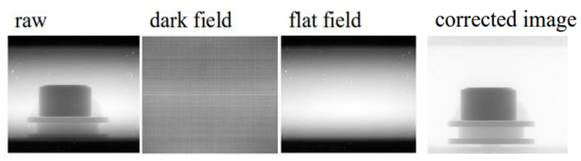


Brightness levels in operations window

Additionally, I explored and contributed to the **flat fielding functionality**, gaining an understanding of the underlying image correction process. Flat fielding enhances image quality by compensating for variations in detector sensitivity and illumination.

$$\text{corrected image} = \frac{\text{raw} - \text{dark field}}{\text{light field} - \text{dark field}}$$

This correction ensures more accurate data representation by normalizing each pixel based on reference images (flat and dark fields), helping to eliminate artifacts and uneven illumination in the final image.

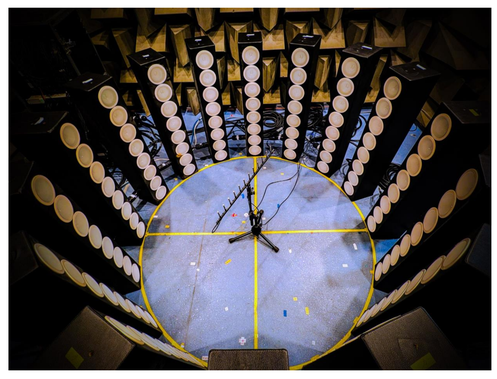


Example of before and after the application of flat fielding

RAL Space

As part of my work with RAL Space, I contributed to the development and evaluation of microphone coherence testing methods for **Direct Field Acoustic Noise (DFAN)** testing, a technique used to simulate the extreme acoustic environments experienced during rocket launches. This project was conducted in close collaboration with **Jonathan Hargreaves** at the **University of Salford**, leveraging his expertise in acoustics to refine experimental methodologies and data interpretation. He also provided a MATLAB script which measures coherency.

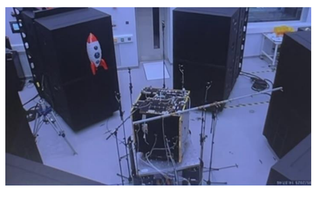
Experimental Setup and Coherence Measurement
Assisted in configuring microphone arrays and conducting tests designed to measure **spatial coherence (γ^2)** between microphone pairs. These measurements are crucial in verifying whether the sound field being generated by DFAN systems is sufficiently diffuse and uniform.



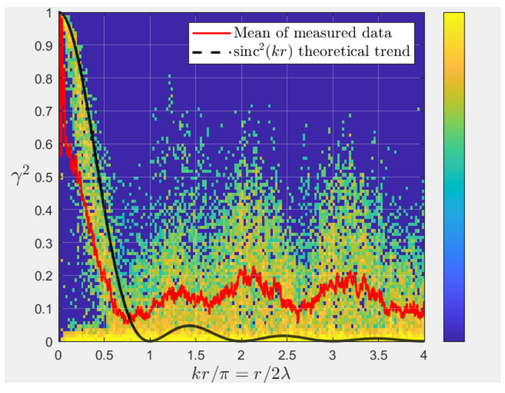
Coherence test set up at University of Salford (Hargreaves et.al)

Development of *Amplify* Software Tool
A Python-based application that automated and streamlined the testing process. The software allowed users to:
•Interface with microphones and data acquisition hardware.
•Collect and visualize live acoustic data during DFAN testing.
•Compute and plot **real-time coherence graphs** across microphone pairs.

The script analyses coherence as a function of normalized distance ($kr/\pi = r/2\lambda$) and compare results with the theoretical $\boldsymbol{sinc^2(kr)}$ curve. This comparison provided a validation benchmark for experimental data and helped assess the quality of the acoustic field.



Coherence test set up at RAL Space



Shows the distribution of measured coherence γ^2

Science Communication

• Volunteered as a tour guide for the synchrotron, hosting Master's students from University of Bristol
• Participated in public engagement events like Stargazing Day, welcoming hundreds of families to campus
• Delivered a technical seminar at Salford University on the importance of computing in scientific research
• Spoke to more than 1500 Y9-11 students as part of the "AMSP Which career?" panel event with Mathematics in Education and Industry

Acknowledgements

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Find out more about ongoing work at ISIS